

Kyu Hyun Lee

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<https://khl02007.github.io/blog/>

EDUCATION

Caltech Ph.D., Biochemistry and Molecular Biophysics option Advisor: Markus Meister	2013 - 2020
Pomona College B.A., Chemistry (minor: Mathematics), magna cum laude	2007 - 2011

EMPLOYMENT

Postdoctoral scholar, UCSF Advisor: Loren Frank	2020 - Current
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RESEARCH INTERESTS

Development and application of novel electrode technology, including flexible neural probes; cognitive influence on visual processing; coordination across hippocampal-cortical circuits; open and reproducible science

SELECTED PUBLICATIONS

Visit my Google Scholar profile for a complete list.

1. Meister, M., **Lee, K.**, and Portugues, R. (2025) *Mathematics in Biology*. MIT Press.
2. **Lee, K.***, Denovellis, E.*, Ly, R., Magland, J., Soules, J., Comrie, A.E., Gramling, D.P., Guidera, J.A., Nevers, R., Adenekan, P., Brozdowski, C., Bray, S., Monroe, E., Bak, J., Coulter, M., Sun, X., Tritt, A., Rübél, O., Nguyen, T., Yatsenko, D., Chu, J., Kemere, C., Garcia, S., Buccino, A., and Frank, L.M. (2024) Spyglass: a data analysis framework for reproducible and shareable neuroscience research. *bioRxiv*. DOI: 10.1101/2024.01.25.577295.
3. **Lee, K.***, Ni, Y.*, Colonell, J., Karsh, B., Putzeys, J., Pachitariu, M., Harris, T.D. and Meister, M. (2021) Electrode pooling can boost the yield of extracellular recordings with switchable silicon probes. *Nature Communications*. 12: 5245.
4. **Lee, K.**, Tran, A., Turan, Z., and Meister, M. (2020) The sifting of visual information in the superior colliculus. *eLife*. 9: e50678.

* denotes equal contribution.

FELLOWSHIPS AND GRANTS

NIH Pathway to Independence Award (K99/R00) (2025 - Current)
Helen Hay Whitney Postdoctoral Fellowship (2022 - 2025)
Tianqiao and Chrissy Chen Institute Graduate Innovator Grant (2018)
Caltech-Amgen Graduate Fellowship (2014 - 2015)

HONORS AND AWARDS

Phi Beta Kappa (2011)
The Robert Rowan III Memorial Award (2011)

PATENTS AND INVENTIONS

Meister, M., **Lee., K.**, and Ni, Y. (2024) Methods and systems for electrode pooling. U.S. Patent 12097033.

INVITED TALKS

Navigating Educational Trajectories in Neuroscience (NIH R25 program), University of Oregon. 2024, 2025.
Dynamic Brain Clusters Databinge Talk, University of British Columbia Center for Brain Health. 2024.
BRAIN Initiative Meeting: Reproducible Workflows for Collaborative Neuroscience symposium, NIH. 2023.
Spike Sorting Workshop, Janelia Research Campus. 2018.
Junior Scientist Workshop on Neural Circuits and Behavior, Janelia Research Campus. 2017.

WORKSHOPS ATTENDED

NWB Developer Days, Janelia Research Campus. 2023.
NeuroData ReHack, Allen Institute for Brain Science. 2022.
Spike Sorting Hackathon, Center for Computational Biology, Flatiron Institute. 2022.
Mountainsort User Workshop, Center for Computational Biology, Flatiron Institute. 2018.

TEACHING EXPERIENCE

Division of Biology and Biological Engineering, Caltech:

- Principles of Neuroscience (Fall term, 2018 - 2019)
- Tools of Neurobiology (Fall term, 2016 - 2017)
- Mathematics in Biology (Fall term, 2015 - 2016)
- Brain Circuits (Winter term, 2014 - 2015)